

popquiz Newton&cie I

⚠ This is a preview of the published version of the quiz

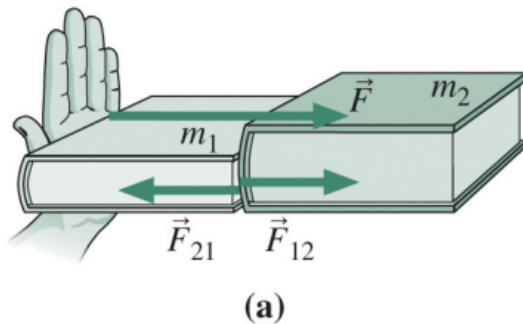
Started: Apr 29 at 7:33pm

Quiz Instructions

for every action there is an opposite and equation reaction. but acceleration depends on the mass



Question 1 17 pts



$m_1 = 1\text{ kg}$, $m_2 = 3\text{ kg}$, the applied force is 50 N . Find the acceleration ?

hint: place both books in an imaginary bag and only consider the external force. ("whats touching the bag")

!!! you want only external force applied to the system. here the "bag" with 2 masses. Use $F=ma$



17 m/s/s



50 m/s/s



12.5 m/s/s



0



Question 2 17 pts

Building on the previous question. Find F_{12} (force applied by the 1kg book on the 3kg book).

hint: now place the 3kg book in an imaginary bag. What is the only force applied to it? the acceleration is the same as previously computed (the books have the same acceleration).

$$F_{12} = m a$$



37.5 N



12.5N



can not be dtermined



50N



Question 3 17 pts

Building on the previous question. The force F_{21} has a magnitude of

hint: for every action you have an equal and opposite reaction



50N



37.5N



12.5N



0N



Question 4 17 pts

the towel trick. Newton's third law at work.



Daniel cormier pushes (discretely) on the towel. Each hand pushes with 0.6 lbs. So the towel pushes back up.

Daniel's real weight is 206.2 lbs. What was his apparent weight with this trick? (the scale reading)

the system is DAniel. he is acted by his weight and by 2 normal force from towel

<https://www.mmafighting.com/2022/7/1/23190921/daniel-cormier-comes-clean-about-towel-gate-during-ufc-hall-of-fame-induction-ceremony>

☐

205.6 lbs

☐

205 lbs

☐

206.2 lbs

☐

207.4

⋮

Question 5 10 pts

Your weight is 150 pounds (action). What is the reaction force?

(hint: what is pulling on you ! I talked about it in class)

☐

The ground is suporting you

☐

there is no reaction force



I spaced out in class



you are pulling on Earth with a force of 150 lbs



Question 6 5 pts

A truck crashes with a mosquito. What is true



not enough information to determine



both the truck and the mosquito experience the same force



the mosquito experiences the largest force



the truck experiences the largest force



Question 7 10 pts

Building on the same question. Which one experiences the largest acceleration



not enough information to determine



both experience the same acceleration



the truck



the mosquito



Question 8 10 pts

Consider a 33 rpm vinyl record. (radius was 15cm) What is true about V (linear velocity) and w (rotational speed)



- ☐ at the center $V=1.57$ cm/s and $w = 0$
- ☐ at the center $V=0$, $w=3.5$ rad/s
- ☐ at the center $V=0$ and $w=0$
- ☐ at the center 0 and $w = 33$ rad/s

Not saved

Submit Quiz